

BUZZARDS BAY, MA, USA

WMO: 994140

Lat: 41.397N

Lon: 71.033W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 12.3	(c) 16.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 46.5	(k) 34.0	(l) 43.3	(m) 32.6	(n) 24.7	(o) 330	(p) 0.795

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.8	(c) 76.5	(d) N/A	(e) 74.9	(f) N/A	(g) 73.6	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 13.6	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 45.1	(b) 39.2	(c) 34.3	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	32.3	34.1	38.1	45.8	53.9	62.3	69.4	69.5	65.0	56.7	47.3	38.9		
	DBStd	14.26	9.48	6.75	5.90	4.97	4.65	4.77	3.37	3.09	3.68	5.86	7.07	7.22		
	HDD50	2013	549	445	368	143	11	0	0	0	0	16	131	350		
	HDD65	5392	1013	865	832	577	344	106	4	3	44	262	531	811		
	CDD50	2454	1	0	1	16	133	369	600	605	451	222	51	5		
	CDD65	354	0	0	0	0	1	25	139	142	44	3	0	0		
	CDH74	247	0	0	0	0	0	10	147	83	7	0	0	0		
	CDH80	12	0	0	0	0	0	0	11	1	0	0	0	0		
(14) Wind	WSAvg	17.7	20.3	20.4	20.2	18.6	17.3	14.6	13.8	13.3	15.1	18.7	20.0	20.6		
(15) Precipitation	PrecAvg	47.5	3.8	3.5	4.6	4.1	3.5	3.8	3.4	4.0	4.2	4.2	3.9	4.6		
	PrecMax	57.0	6.8	7.1	14.4	8.0	7.5	11.3	9.9	7.9	9.2	13.9	6.4	7.8		
	PrecMin	37.1	1.7	1.7	0.3	1.4	1.1	0.1	1.4	1.5	1.2	0.6	0.8	1.3		
	PrecStd	5.9	1.3	1.3	2.8	1.8	1.4	2.8	1.6	1.8	2.0	2.8	1.5	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	49.8	54.1	64.3	70.9	75.3	81.5	78.4	75.2	70.4	62.7	57.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	50.7	47.4	50.5	57.0	65.6	72.0	77.4	76.0	72.6	67.9	60.3	54.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	48.1	45.0	48.0	53.7	62.6	70.2	75.3	74.8	71.0	66.0	59.0	52.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	45.2	42.8	46.0	51.5	60.2	68.3	73.9	73.8	69.6	64.2	57.5	49.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	9.4	8.5	7.4	6.9	6.2	5.3	4.8	5.8	7.1	8.4	9.5		
		MCDBR	11.8	11.4	10.7	11.4	11.3	9.1	7.8	6.0	6.7	7.4	8.4	11.3		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.311	0.319	0.342	0.378	0.420	0.465	0.492	0.446	0.379	0.350	0.321	0.309		
	taud		2.502	2.483	2.452	2.382	2.302	2.215	2.149	2.271	2.434	2.515	2.556	2.552		
	Ebn at Noon		268	284	288	283	272	259	251	259	272	268	263	260		
	Edh at Noon		23	28	32	37	41	45	47	41	32	27	22	21		
(44) All-Sky Solar Radiation	RadAvg		580	864	1192	1539	1715	1864	1912	1719	1401	953	645	490		
	RadStd		46	86	90	164	165	180	102	94	103	84	46	48		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+176	+88	

Nomenclature: See separate page